JIAZHENG SONG

Email: jiazhengsong@zju.edu.cn |

PERSONAL PROFILE

I am a second year Ph.D. student in Electronic Information at Zhejiang University. My research interests focus on data-centric methods for constructing efficient, reliable, and socially beneficial agents.

EDUCATION

Zhejiang University 09/06/2024 – Present

Ph.D. Student in Electronic Information

- **Core Courses:** Knowledge Graph, Mathematical Modeling, Information and Computing

Zhejiang University 09/01/2020 – 06/30/2024

Bachelor's Degree (Computer Science)

- **Core Courses:** Operating Systems, Database Systems, Computer Architecture, Optimization Theory.
- **Honors and Awards:**
  - Outstanding Graduates of Zhejiang University / Outstanding Undergraduate Thesis Award: for the thesis titled "Shapley Value Calculation Based on Multiplicative Marginal Contribution".

PUBLICATION

ASSS: Adaptive Stratified Sampling for Shapley-like Values J. Pang, J. Zhang, **Jiazheng Song**, J. Liu, K. Ren. *Proceedings of the ACM on Management of Data*, Volume 4, Issue 3, Article 139, May 2026

Summary: This paper proposes adaptive stratified sampling strategies (ASSS) to estimate Shapley-like values efficiently. Unlike static stratified sampling that requires $\Omega(n^2 \log n)$ samples, ASSS retains unbiasedness with fewer samples by adaptively adjusting the number of strata. Two approaches, ASSS-A (stratify after sampling) and ASSS-B (stratify before sampling), offer flexible trade-offs between sample efficiency and memory usage. Experiments on real and synthetic datasets demonstrate the effectiveness of the proposed methods.

RESEARCH EXPERIENCE

Representative Functional Dependencies Discovery

Description: This work formulated the problem of discovering a representative subset of functional dependencies (FDs) to avoid duplicative information, as the number of FDs grows drastically with dataset dimensionality. Instead of selecting representatives after full discovery, we integrated representativeness verification directly into the lattice traversal process, validating only representative FD candidates. The algorithm was enhanced with nearest indexes for search space pruning, dynamic stripped partitions to accelerate FD validation, and a compressed FD-tree to guarantee FD minimality.

Conference: Accepted to IEEE ICDE 2026, presented in Montreal, Canada on May 12, 2026.

SafeAdapt: Safety Alignment with Adaptive Thinking Allocation for Large Reasoning Models

Description: This work systematically evaluated the relationship between reasoning budget and model behavioral robustness in large reasoning models under adversarial conditions. The key finding was that behavior does not monotonically improve with longer thinking trajectories; both over-thinking and under-thinking cause robustness degradation. Based on this, I proposed SafeAdapt, which dynamically adjusts thinking allocation according to prompt difficulty, enabling adaptive resource allocation instead of a fixed pattern. Experiments on multiple adversarial benchmarks showed significant improvement in behavioral robustness with a more ideal thinking budget.

Conference: Accepted to USENIX Security Symposium 2026; to be presented in Baltimore, USA on August 12–14, 2026.